

Poroviscoelastic Modelling of Brain Oedema

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Introduction

Precision in neurosurgical interventions, such as Deep Brain Stimulation (DBS), is often compromised by brain shift: intra-operative deformation of brain tissue caused by factors such as cerebrospinal fluid (CSF) loss, changes in intracranial pressure (ICP), or oedema development. Brain oedema, defined as fluid accumulation and tissue swelling, is a common pathological response in traumatic brain injury, stroke, and tumours. Since the brain, cerebral blood, and CSF are enclosed within the rigid skull, even small volume changes may substantially increase ICP and affect cerebral blood flow [1, 2].

These deformations motivate biomechanical simulations able to provide patient-specific computational predictions. Brain tissue is a biphasic, ultrasoft material composed mainly of fluid and a solid matrix. Its response combines viscoelasticity, associated with solid matrix rearrangement, and poroelasticity, associated with fluid migration through the porous tissue. Viscoelastic effects dominate short-term behaviour, whereas poroelastic relaxation becomes more relevant over longer time scales; therefore, accurate models must account for nonlinear, region-dependent, and coupled porous-viscous effects [3, 4, 5].

This practical work develops a multilayer biphasic Finite Element (FE) model to simulate oedema-induced brain deformation. The brain is represented as a heterogeneous structure with four anatomical regions: cortex (C), basal ganglia (BG), corona radiata (CR), and corpus callosum (CC), as shown in Figure 1. Using a poroviscoelastic framework, the simulation evaluates tissue displacement and mechanical state under incremental oedema pressure over 72 hours, combining Ogden-type hyperelastic properties with time-dependent viscous parameters [6].

Methods

The mechanical response of brain tissue to oedema-induced pressure was analysed with the Finite Element Method (FEM) using FEBio v4.12.0. A transient, large-deformation biphasic analysis was implemented to solve the coupled problem of solid matrix deformation and interstitial fluid migration. The simulation converged over a 72-hour period, discretised into 500 time steps of 518.4 s, using a mesh with 14,436 nodes and 13,392 solid elements [6].

The brain was approximated as a multilayer heterogeneous structure with an overall diameter of 150 mm. Four concentric anatomical regions were represented from the inside to the outside: basal ganglia (BG), represented as the innermost subcortical region with a diameter of 20 mm; corpus callosum (CC), an internal white-matter structure with a thickness of 6 mm; corona radiata (CR), an intermediate white-matter region with a thickness of 56 mm; and cortex (C), modelled as the

outermost 3 mm layer.

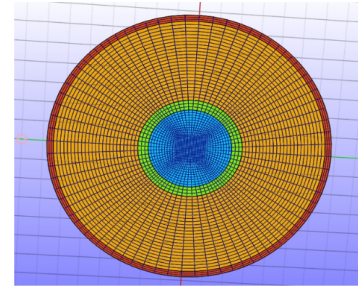


Figure 1: Multilayer geometrical approximation of the brain model, ordered from the centre outwards as basal ganglia, corpus callosum, corona radiata, and cortex.

Brain tissue was described as a poroviscoelastic material to account for its biphasic composition, approximately 90% fluid and 10% solid matrix. The material calibration was based on the regional Ogden parameters and viscoelastic relaxation parameters reported in Tables 6 and 7 of Budday et al. [5]. These source values were converted into the parameters required by FEBio, adapting the original constants to the FEBio formulation and unit system before assigning them to each anatomical region.

The solid phase was modelled using an Ogden hyperelastic law, suitable for the nonlinear ultrasoft response of neural tissue and consistent with the nonlinear pre-conditioning behaviour shown in Figure 2. The strain-energy density function was

$$W = \sum_{p=1}^N \frac{\mu_p}{\alpha_p^2} \left(\lambda_1^{\alpha_p} + \lambda_2^{\alpha_p} + \lambda_3^{\alpha_p} - 3 \right), \quad (1)$$

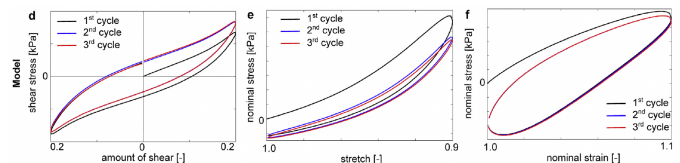


Figure 2: Predicted pre-conditioning behaviour in shear, compression, and tension from the viscoelastic constitutive model, corresponding to panels d–f of the original figure from Budday et al. [5].

where μ_p and α_p are material constants and λ_i are the principal stretches. For FEBio, the long-term shear parameter μ_∞ from Budday et al. [5] was converted to the Ogden coefficient $c_1 = 2\mu_\infty$, while the Ogden exponent was kept as $m_1 = \alpha$. Time-dependent stress relaxation of the matrix was incorporated through a Prony-series viscoelastic formulation. The two Prony coefficients required by FEBio were obtained by combining the fitted viscous parameters from Table 6 with

the characteristic relaxation times from Table 7 of Budday et al. For each Maxwell branch, the non-equilibrium shear modulus was first recovered as viscosity divided by relaxation time, $\mu_1 = \theta_1/\tau_1$ and $\mu_2 = \theta_2/\tau_2$. These moduli were then normalised by the long-term shear modulus to obtain the dimensionless FEBio coefficients, $G_1 = \mu_1/\mu_\infty = (\theta_1/\tau_1)/\mu_\infty$ and $G_2 = \mu_2/\mu_\infty = (\theta_2/\tau_2)/\mu_\infty$. Hydraulic permeability was taken from Su et al. [3] as $2.42 \times 10^{-14} \text{ m}^4/\text{Ns}$ and converted to FEBio units as $2.42 \times 10^{-2} \text{ mm}^4/\text{Ns}$. The final FEBio parameters are summarised in Table 1.

Table 1. FEBio poroviscoelastic material parameters computed from the source Ogden and viscoelastic data.

Region	c_1 (MPa)	m_1	G_1	G_2
C	8.40×10^{-4}	-21.27	3.33	1.33
BG	3.40×10^{-4}	-21.25	4.00	1.59
CR	3.20×10^{-4}	-25.66	6.06	1.56
CC	8.00×10^{-5}	-28.41	15.76	4.00

Permeability was assumed isotropic and homogeneous across regions because a single literature value was used for the whole model. This simplification is consistent with the aim of capturing long-term poroelastic relaxation over hours to days, while the Prony terms describe the short-term viscoelastic response. To mimic oedema development inside the rigid skull, the outer cortical boundary was fixed and the innermost basal-ganglia region was constrained as a central anchor. A fluid pressure load was then applied incrementally from 0 to 1 kPa over 72 hours at the nodes assigned to oedema fluid pressure. The 72-hour loading window and final pressure level were chosen to represent oedema growth towards a pathological ICP of approximately 30 mmHg, following the pressure-volume relationship illustrated by Bothwell et al. [7].

Results and Discussion

The FEBio simulation ran until normal termination, covering the full 72-hour oedema period, or 259,000 s of simulated time. The solver completed the analysis with stable convergence, meaning that the coupled biphasic problem was solved successfully without numerical failure.

The mechanical response was progressive rather than abrupt. As oedema pressure increased, two corpus callosum elements close to the oedema region were monitored (Figure 3). Element E4629 was located on the inner side of the corpus callosum, in contact with the basal ganglia, whereas E5506 was located on its outer side, in contact with the corona radiata. Both points showed a monotonic rise in effective stress (Figure 4), but E4629 reached the largest value, approximately $2.4 \times 10^{-3} \text{ MPa}$, whereas E5506 remained lower, around $5.2 \times 10^{-4} \text{ MPa}$. This spatial difference suggests stress concentration towards the inner corpus-callosum boundary, closer to the constrained and pressurised basal-ganglia region, while surrounding tissue redistributed part of the load through poroelastic fluid migration and viscoelastic relaxation.

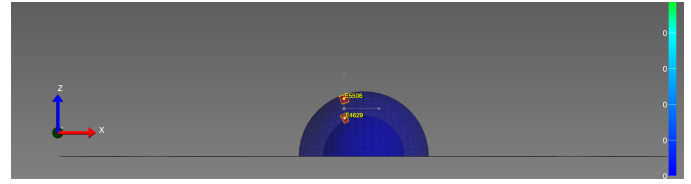


Figure 3: Location of the monitored corpus callosum elements close to the oedema region. E4629 is located on the inner side, in contact with the basal ganglia, and E5506 on the outer side, in contact with the corona radiata.

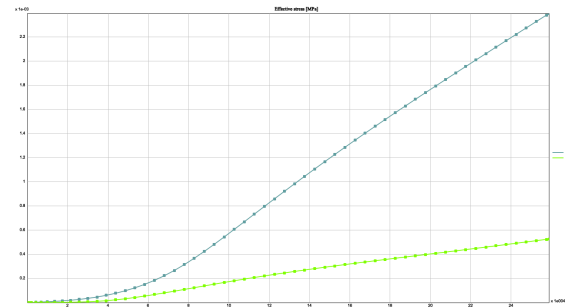


Figure 4: Effective-stress response of the monitored corpus callosum elements during the 72-hour oedema simulation.

The final displacement field was concentrated around the oedema origin in the cut model, with a maximum of approximately 3.18 mm (Figures 5 and 6). The accompanying fluid-flux vectors radiated from the oedema centre, supporting the interpretation that pressure-driven fluid migration was the main driver of the local deformation pattern.

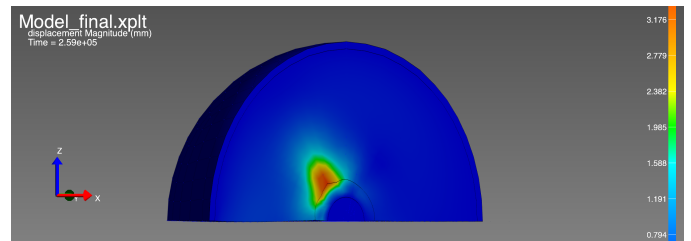


Figure 5: Final displacement magnitude at 72 hours in the cut model, showing deformation concentrated around the oedema source.

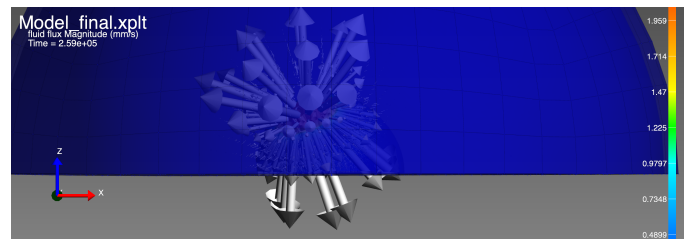


Figure 6: Final fluid-flux magnitude and vectors at 72 hours, radiating from the oedema centre.

These results are consistent with the workshop modelling assumptions: the rigid skull and fixed internal anchor restrict deformation, whereas the biphasic material allows pressure-induced fluid redistribution over long time scales. The response therefore reflects both mechanical confinement and

tissue permeability. However, the model remains simplified because the geometry is layered, permeability is homogeneous and isotropic, and the oedema load is prescribed rather than patient-specific.

In conclusion, the model provides a useful first approximation of how oedema-induced pressure can produce region-dependent mechanical stress in brain tissue. Such simulations could support neurosurgical planning or sensitivity studies for oedema progression, but further development would require patient-specific anatomy, anisotropic white-matter properties, improved boundary conditions, and validation against clinical or experimental deformation data.

References

- [1] V. Wykes and R. Vindlacheruvu. Intracranial pressure, cerebral blood flow and brain oedema. *Neurosurgery*, 33(3), 2012.
- [2] A. Hewitt and C. Ellory. Brain oedema, intracranial pressure and cerebral blood flow. *Surgery*, 30(3), 2012.
- [3] L. Su et al. Distinguishing poroelasticity and viscoelasticity of brain tissue with time scale. *Acta Biomaterialia*, 155, 2023.
- [4] E. Comellas et al. Modeling the porous and viscous responses of human brain tissue behavior. *Computer Methods in Applied Mechanics and Engineering*, 369, 2020.
- [5] S. Budday et al. Rheological characterization of human brain tissue. *Acta Biomaterialia*, 60, 2017.
- [6] J. Noailly. Brain modelling. *Computational Therapies Course Material*, Universitat Pompeu Fabra.
- [7] S. W. Bothwell, D. Janigro, and A. Patabendige. Cerebrospinal fluid dynamics and intracranial pressure elevation in neurological diseases. *Fluids and Barriers of the CNS*, 16(9), 2019.